

assignment_7_solutions_and_rubric

November 27, 2023

```
[1]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from matplotlib import colors
```

0.1 Problem #1a: 5 pts for downloading data

0.1.1 rubric:

-5 for not being able to load the data

```
[2]: filename = "PSCompPars_2023.11.20_08.50.11.csv"
```

```
[3]: dat = pd.read_csv(filename, sep=",", skiprows=88,)
```

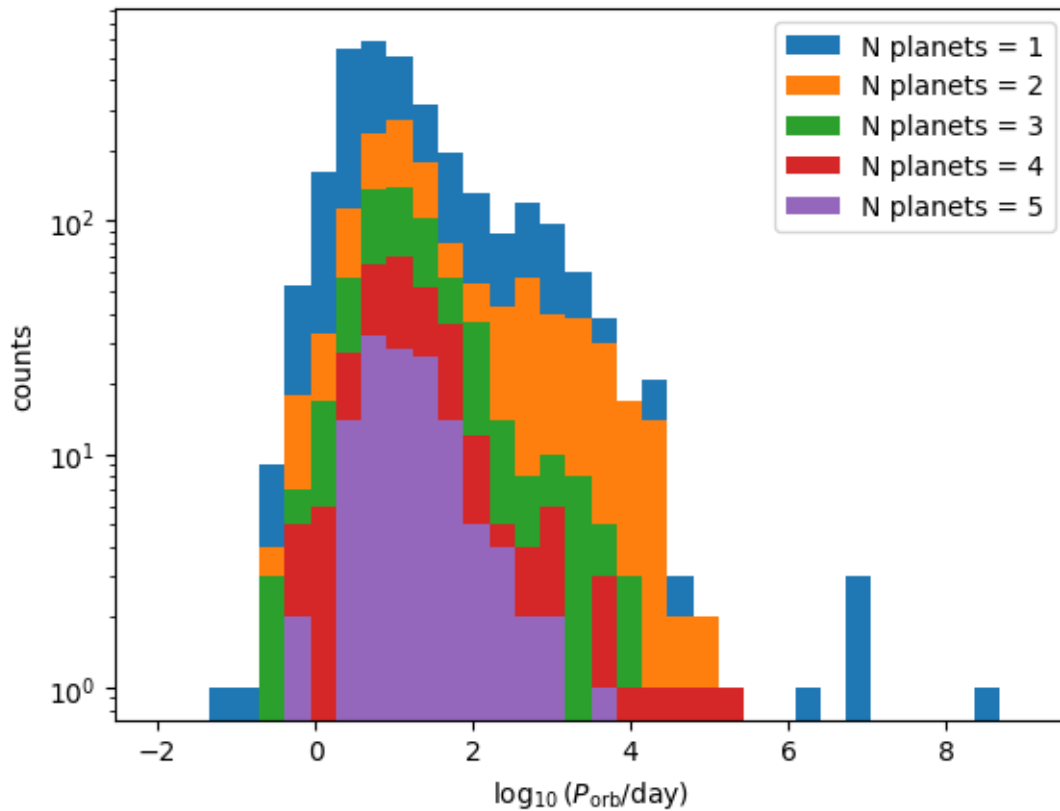
0.2 Problem #1b: 10 pts for the figure with the 5 datasets, labels, and legend

0.2.1 rubric:

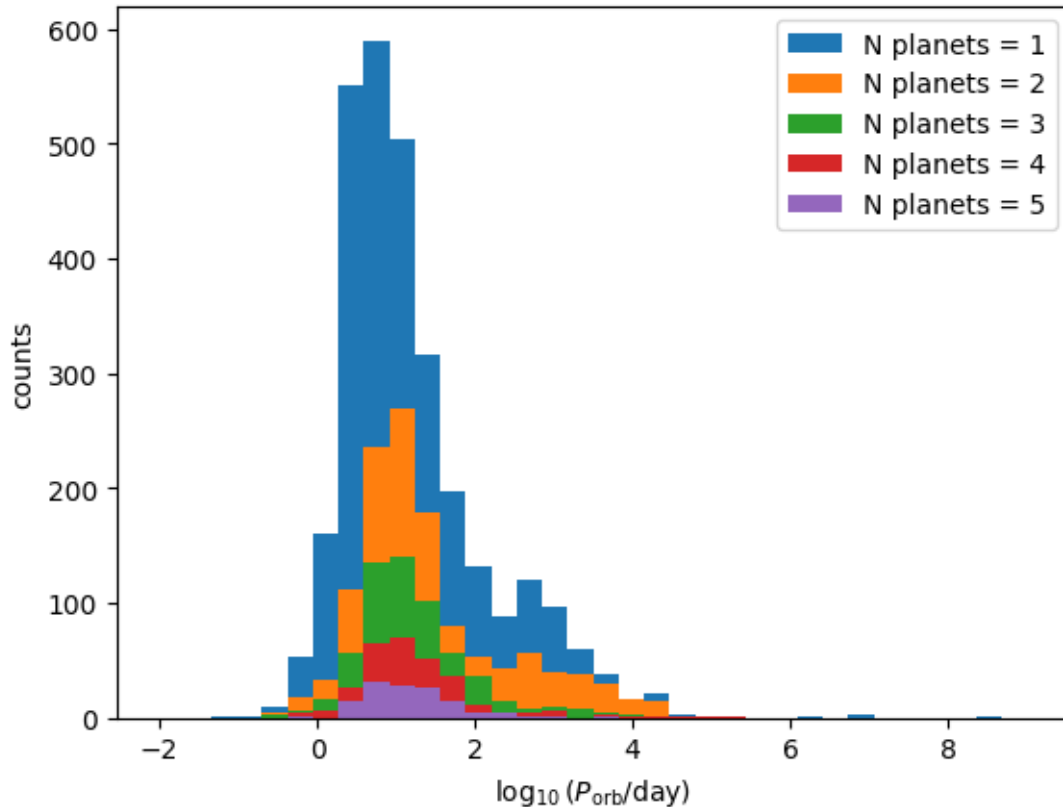
-1 for each missing axis label or legend

-1 for each missing dataset (1, 2, 3, 4, 5 planet systems)

```
[4]: for ii in range(1,6):
    dat_select = dat.loc[dat.sy_pnum == ii]
    plt.hist(np.log10(dat_select.pl_orbper), bins=np.linspace(-2, 9, 35),
             label=f"N planets = {ii}", density=False)
plt.legend()
plt.yscale('log')
plt.xlabel(r'$\log_{10}(P_{\rm orb}$/day)')
plt.ylabel('counts')
plt.show()
```



```
[5]: for ii in range(1,6):
      dat_select = dat.loc[dat.sy_pnum == ii]
      plt.hist(np.log10(dat_select.pl_orbper), bins=np.linspace(-2, 9, 35),
               label=f"N planets = {ii}", density=False)
      plt.legend()
      plt.xlabel(r'$\log_{10}(P_{\text{orb}}/\text{day})$')
      plt.ylabel('counts')
      plt.show()
```



0.3 problem #1c:

observational biases (3 pts each; 6 pts total for 2 biases)

- 1) The transit method and radial velocity methods both are likely to observe transits at shorter periods, hence the peak in the distribution below 100 days for all the cases
- 2) Finding multiplanet systems usually requires followup observations (like through transit timing variations) that are expensive to obtain, so you are more likely to find more single planet systems based on the current survey strategy
- 3) Very fast rotating stars have starspots that transit on day-ish timescales, so it is difficult to confirm planets at those short periods

exoplanet dynamics (3 pts for 1 reason)

- 1) Secular resonances occur at fixed orbital period ratios and thus as more planets are added to the system, more short period systems (at higher period ratios) are required
- 2) Very wide orbits of exoplanets can be disrupted due to fly by's of stellar objects

0.3.1 rubric:

-1.5 for unreasonable answer

-3 for each missing bias or dynamics explanation

0.4 Problem 2: 20 pts (10 pts for figure, 10 pts for caption)

0.4.1 rubric:

-2 each for missing axis labels or legend

-2 for missing punchline

-2 for not describing figure accurately, or for not explaining specific trends in data

[]: